STRAVA FITNESS
My SQL Report

SQL WORKFLOW:

Tool used: **MySQL Work Bench**

```
--  ================================= --
-- |       **STRAVA FITNESS**        |
--  ================================= --
```

Database Initialization:

```
Create database Strava;
use Strava;
```

LOAD ALL THE DATASETS

1. dailyactivity

```
CREATE TABLE dailyactivity (
    Id BIGINT,
    ActivityDate DATE,
    TotalSteps INT,
    TotalDistance FLOAT,
    TrackerDistance FLOAT,
    LoggedActivitiesDistance FLOAT,
    VeryActiveDistance FLOAT,
    ModeratelyActiveDistance FLOAT,
    LightActiveDistance FLOAT,
    SedentaryActiveDistance FLOAT,
    VeryActiveMinutes INT,
    FairlyActiveMinutes INT,
    LightlyActiveMinutes INT,
    SedentaryMinutes INT,
    Calories INT
);
SET GLOBAL local_infile = 1;

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyActivity_merged.csv"
INTO TABLE dailyactivity
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDate, @TotalSteps, @TotalDistance, @TrackerDistance,
@LoggedActivitiesDistance,
 @VeryActiveDistance, @ModeratelyActiveDistance, @LightActiveDistance,
@SedentaryActiveDistance,
 @VeryActiveMinutes, @FairlyActiveMinutes, @LightlyActiveMinutes,
@SedentaryMinutes, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityDate = STR_TO_DATE(@ActivityDate, '%c/%e/%Y'),  -- Handles
single-digit months/days
    TotalSteps = NULLIF(@TotalSteps, ''),
    TotalDistance = NULLIF(@TotalDistance, ''),
    TrackerDistance = NULLIF(@TrackerDistance, ''),
```

```
    LoggedActivitiesDistance = NULLIF(@LoggedActivitiesDistance, ''),
    VeryActiveDistance = NULLIF(@VeryActiveDistance, ''),
    ModeratelyActiveDistance = NULLIF(@ModeratelyActiveDistance, ''),
    LightActiveDistance = NULLIF(@LightActiveDistance, ''),
    SedentaryActiveDistance = NULLIF(@SedentaryActiveDistance, ''),
    VeryActiveMinutes = NULLIF(@VeryActiveMinutes, ''),
    FairlyActiveMinutes = NULLIF(@FairlyActiveMinutes, ''),
    LightlyActiveMinutes = NULLIF(@LightlyActiveMinutes, ''),
    SedentaryMinutes = NULLIF(@SedentaryMinutes, ''),
    Calories = NULLIF(@Calories, '');
```

2. dailycalories:

```
CREATE TABLE dailycalories (
    Id BIGINT,
    ActivityDay DATE,
    Calories INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyCalories_merged.csv"
INTO TABLE dailycalories
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
    Calories = NULLIF(@Calories, '');
```

3. dailyintensities

```
CREATE TABLE dailyintensities (
    Id BIGINT,
    ActivityDay DATE,
    SedentaryMinutes INT,
    LightlyActiveMinutes INT,
    FairlyActiveMinutes INT,
    VeryActiveMinutes INT,
    SedentaryActiveDistance int,
      LightActiveDistance double,
    ModeratelyActiveDistance double,
    VeryActiveDistance double
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyIntensities_merged.csv"
INTO TABLE dailyintensities
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @SedentaryMinutes, @LightlyActiveMinutes,
@FairlyActiveMinutes, @VeryActiveMinutes, @SedentaryActiveDistance,
@LightActiveDistance,@ModeratelyActiveDistance,@VeryActiveDistance)
SET
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
```

```
    SedentaryMinutes = NULLIF(@SedentaryMinutes, ''),
    LightlyActiveMinutes = NULLIF(@LightlyActiveMinutes, ''),
    FairlyActiveMinutes = NULLIF(@FairlyActiveMinutes, ''),
    VeryActiveMinutes = NULLIF(@VeryActiveMinutes, ''),
    SedentaryActiveDistance = NULLIF(@SedentaryActiveDistance, ''),
    LightActiveDistance = NULLIF(@LightActiveDistance, ''),
    ModeratelyActiveDistance = NULLIF(@ModeratelyActiveDistance, ''),
    VeryActiveDistance = NULLIF(@VeryActiveDistance, '');
```

4. Dailysteps

```
CREATE TABLE dailysteps (
    Id BIGINT,
    ActivityDay DATE,
    StepTotal INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailySteps_merged.csv"
INTO TABLE dailysteps
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @StepTotal)
SET
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
    StepTotal = NULLIF(@StepTotal, '');
```

5. Heartrateseconds

```
CREATE TABLE heartrateseconds (
    Id BIGINT,
    Time DATETIME,
    Value INT);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/heartrate_seconds_merged.csv"
INTO TABLE heartrateseconds
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Time, @Value)
SET
    Id = NULLIF(@Id, ''),
    Time = STR_TO_DATE(@Time, '%m/%d/%Y %r'),
    Value = NULLIF(@Value, '');
```

6. Hourlycalories

```
CREATE TABLE hourlycalories (
    Id BIGINT,
    ActivityHour DATETIME,
    Calories INT
);
```

```
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlyCalories_merged.csv"
INTO TABLE hourlycalories
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Calories = NULLIF(@Calories, '');
```

7. Hourlyintensities

```
 CREATE TABLE hourlyintensities (
    Id BIGINT,
    ActivityHour DATETIME,
    TotalIntensity INT,
    AverageIntensity DOUBLE
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlyIntensities_merged.csv"
INTO TABLE hourlyintensities
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @TotalIntensity, @AverageIntensity)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    TotalIntensity = NULLIF(@TotalIntensity, ''),
    AverageIntensity = NULLIF(@AverageIntensity, '');
```

8. hourlysteps

```
CREATE TABLE hourlysteps (
    Id BIGINT,
    ActivityHour DATETIME,
    StepTotal INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlySteps_merged.csv"
INTO TABLE hourlysteps
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @StepTotal)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    StepTotal = NULLIF(@StepTotal, '');
```

9. Minutecaloriesnarrow

```
 CREATE TABLE minutecaloriesnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    Calories FLOAT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteCaloriesNarrow_merged.csv"
INTO TABLE minutecaloriesnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Calories = NULLIF(@Calories, '');
```

10. Minutecalorieswide

```
CREATE TABLE minutecalorieswide (
    Id BIGINT,
    ActivityMinute DATETIME,
    Calories00 DOUBLE,
    Calories01 DOUBLE,
    Calories02 DOUBLE,
    Calories03 DOUBLE,
    Calories04 DOUBLE,
    Calories05 DOUBLE,
    Calories06 DOUBLE,
    Calories07 DOUBLE,
    Calories08 DOUBLE,
    Calories09 DOUBLE,
    Calories10 DOUBLE,
    Calories11 DOUBLE,
    Calories12 DOUBLE,
    Calories13 DOUBLE,
    Calories14 DOUBLE,
    Calories15 DOUBLE,
    Calories16 DOUBLE,
    Calories17 DOUBLE,
    Calories18 DOUBLE,
    Calories19 DOUBLE,
    Calories20 DOUBLE,
    Calories21 DOUBLE,
    Calories22 DOUBLE,
    Calories23 DOUBLE,
    Calories24 DOUBLE,
    Calories25 DOUBLE,
    Calories26 DOUBLE,
    Calories27 DOUBLE,
    Calories28 DOUBLE,
    Calories29 DOUBLE,
    Calories30 DOUBLE,
    Calories31 DOUBLE,
    Calories32 DOUBLE,
```

```
    Calories33 DOUBLE,
    Calories34 DOUBLE,
    Calories35 DOUBLE,
    Calories36 DOUBLE,
    Calories37 DOUBLE,
    Calories38 DOUBLE,
    Calories39 DOUBLE,
    Calories40 DOUBLE,
    Calories41 DOUBLE,
    Calories42 DOUBLE,
    Calories43 DOUBLE,
    Calories44 DOUBLE,
    Calories45 DOUBLE,
    Calories46 DOUBLE,
    Calories47 DOUBLE,
    Calories48 DOUBLE,
    Calories49 DOUBLE,
    Calories50 DOUBLE,
    Calories51 DOUBLE,
    Calories52 DOUBLE,
    Calories53 DOUBLE,
    Calories54 DOUBLE,
    Calories55 DOUBLE,
    Calories56 DOUBLE,
    Calories57 DOUBLE,
    Calories58 DOUBLE,
    Calories59 DOUBLE
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteCaloriesWide_merged.csv"
INTO TABLE minutecalorieswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS

(@Id, @ActivityMinute,
 @Calories00, @Calories01, @Calories02, @Calories03, @Calories04,
@Calories05, @Calories06, @Calories07, @Calories08, @Calories09,
 @Calories10, @Calories11, @Calories12, @Calories13, @Calories14,
@Calories15, @Calories16, @Calories17, @Calories18, @Calories19,
 @Calories20, @Calories21, @Calories22, @Calories23, @Calories24,
@Calories25, @Calories26, @Calories27, @Calories28, @Calories29,
 @Calories30, @Calories31, @Calories32, @Calories33, @Calories34,
@Calories35, @Calories36, @Calories37, @Calories38, @Calories39,
 @Calories40, @Calories41, @Calories42, @Calories43, @Calories44,
@Calories45, @Calories46, @Calories47, @Calories48, @Calories49,
 @Calories50, @Calories51, @Calories52, @Calories53, @Calories54,
@Calories55, @Calories56, @Calories57, @Calories58, @Calories59)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Calories00 = NULLIF(@Calories00, ''),
    Calories01 = NULLIF(@Calories01, ''),
    Calories02 = NULLIF(@Calories02, ''),
    Calories03 = NULLIF(@Calories03, ''),
    Calories04 = NULLIF(@Calories04, ''),
    Calories05 = NULLIF(@Calories05, ''),
```

```
    Calories06 = NULLIF(@Calories06, ''),
    Calories07 = NULLIF(@Calories07, ''),
    Calories08 = NULLIF(@Calories08, ''),
    Calories09 = NULLIF(@Calories09, ''),
    Calories10 = NULLIF(@Calories10, ''),
    Calories11 = NULLIF(@Calories11, ''),
    Calories12 = NULLIF(@Calories12, ''),
    Calories13 = NULLIF(@Calories13, ''),
    Calories14 = NULLIF(@Calories14, ''),
    Calories15 = NULLIF(@Calories15, ''),
    Calories16 = NULLIF(@Calories16, ''),
    Calories17 = NULLIF(@Calories17, ''),
    Calories18 = NULLIF(@Calories18, ''),
    Calories19 = NULLIF(@Calories19, ''),
    Calories20 = NULLIF(@Calories20, ''),
    Calories21 = NULLIF(@Calories21, ''),
    Calories22 = NULLIF(@Calories22, ''),
    Calories23 = NULLIF(@Calories23, ''),
    Calories24 = NULLIF(@Calories24, ''),
    Calories25 = NULLIF(@Calories25, ''),
    Calories26 = NULLIF(@Calories26, ''),
    Calories27 = NULLIF(@Calories27, ''),
    Calories28 = NULLIF(@Calories28, ''),
    Calories29 = NULLIF(@Calories29, ''),
    Calories30 = NULLIF(@Calories30, ''),
    Calories31 = NULLIF(@Calories31, ''),
    Calories32 = NULLIF(@Calories32, ''),
    Calories33 = NULLIF(@Calories33, ''),
    Calories34 = NULLIF(@Calories34, ''),
    Calories35 = NULLIF(@Calories35, ''),
    Calories36 = NULLIF(@Calories36, ''),
    Calories37 = NULLIF(@Calories37, ''),
    Calories38 = NULLIF(@Calories38, ''),
    Calories39 = NULLIF(@Calories39, ''),
    Calories40 = NULLIF(@Calories40, ''),
    Calories41 = NULLIF(@Calories41, ''),
    Calories42 = NULLIF(@Calories42, ''),
    Calories43 = NULLIF(@Calories43, ''),
    Calories44 = NULLIF(@Calories44, ''),
    Calories45 = NULLIF(@Calories45, ''),
    Calories46 = NULLIF(@Calories46, ''),
    Calories47 = NULLIF(@Calories47, ''),
    Calories48 = NULLIF(@Calories48, ''),
    Calories49 = NULLIF(@Calories49, ''),
    Calories50 = NULLIF(@Calories50, ''),
    Calories51 = NULLIF(@Calories51, ''),
    Calories52 = NULLIF(@Calories52, ''),
    Calories53 = NULLIF(@Calories53, ''),
    Calories54 = NULLIF(@Calories54, ''),
    Calories55 = NULLIF(@Calories55, ''),
    Calories56 = NULLIF(@Calories56, ''),
    Calories57 = NULLIF(@Calories57, ''),
    Calories58 = NULLIF(@Calories58, ''),
    Calories59 = NULLIF(@Calories59, '');
```

11. minuteintensitiesnarrow

```
CREATE TABLE minuteintensitiesnarrow (
```

```
    Id BIGINT,
    ActivityMinute DATETIME,
    Intensity INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteIntensitiesNarrow_merged.csv"
INTO TABLE minuteintensitiesnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Intensity)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Intensity = NULLIF(@Intensity, '');
```

12. ***minuteintensitieswide***

```
CREATE TABLE minuteintensitieswide (
    Id BIGINT,
    ActivityHour DATETIME,
    Intensity00 INT,
    Intensity01 INT,
    Intensity02 INT,
    Intensity03 INT,
    Intensity04 INT,
    Intensity05 INT,
    Intensity06 INT,
    Intensity07 INT,
    Intensity08 INT,
    Intensity09 INT,
    Intensity10 INT,
    Intensity11 INT,
    Intensity12 INT,
    Intensity13 INT,
    Intensity14 INT,
    Intensity15 INT,
    Intensity16 INT,
    Intensity17 INT,
    Intensity18 INT,
    Intensity19 INT,
    Intensity20 INT,
    Intensity21 INT,
    Intensity22 INT,
    Intensity23 INT,
    Intensity24 INT,
    Intensity25 INT,
    Intensity26 INT,
    Intensity27 INT,
    Intensity28 INT,
    Intensity29 INT,
    Intensity30 INT,
    Intensity31 INT,
    Intensity32 INT,
    Intensity33 INT,
    Intensity34 INT,
    Intensity35 INT,
```

```
    Intensity36 INT,
    Intensity37 INT,
    Intensity38 INT,
    Intensity39 INT,
    Intensity40 INT,
    Intensity41 INT,
    Intensity42 INT,
    Intensity43 INT,
    Intensity44 INT,
    Intensity45 INT,
    Intensity46 INT,
    Intensity47 INT,
    Intensity48 INT,
    Intensity49 INT,
    Intensity50 INT,
    Intensity51 INT,
    Intensity52 INT,
    Intensity53 INT,
    Intensity54 INT,
    Intensity55 INT,
    Intensity56 INT,
    Intensity57 INT,
    Intensity58 INT,
    Intensity59 INT
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteIntensitiesWide_merged.csv"
INTO TABLE minuteintensitieswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS

(@Id, @ActivityHour,
 @Intensity00, @Intensity01, @Intensity02, @Intensity03, @Intensity04,
@Intensity05, @Intensity06, @Intensity07, @Intensity08, @Intensity09,
 @Intensity10, @Intensity11, @Intensity12, @Intensity13, @Intensity14,
@Intensity15, @Intensity16, @Intensity17, @Intensity18, @Intensity19,
 @Intensity20, @Intensity21, @Intensity22, @Intensity23, @Intensity24,
@Intensity25, @Intensity26, @Intensity27, @Intensity28, @Intensity29,
 @Intensity30, @Intensity31, @Intensity32, @Intensity33, @Intensity34,
@Intensity35, @Intensity36, @Intensity37, @Intensity38, @Intensity39,
 @Intensity40, @Intensity41, @Intensity42, @Intensity43, @Intensity44,
@Intensity45, @Intensity46, @Intensity47, @Intensity48, @Intensity49,
 @Intensity50, @Intensity51, @Intensity52, @Intensity53, @Intensity54,
@Intensity55, @Intensity56, @Intensity57, @Intensity58, @Intensity59)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Intensity00 = NULLIF(@Intensity00, ''),
    Intensity01 = NULLIF(@Intensity01, ''),
    Intensity02 = NULLIF(@Intensity02, ''),
    Intensity03 = NULLIF(@Intensity03, ''),
    Intensity04 = NULLIF(@Intensity04, ''),
    Intensity05 = NULLIF(@Intensity05, ''),
    Intensity06 = NULLIF(@Intensity06, ''),
    Intensity07 = NULLIF(@Intensity07, ''),
    Intensity08 = NULLIF(@Intensity08, ''),
```

```
    Intensity09 = NULLIF(@Intensity09, ''),
    Intensity10 = NULLIF(@Intensity10, ''),
    Intensity11 = NULLIF(@Intensity11, ''),
    Intensity12 = NULLIF(@Intensity12, ''),
    Intensity13 = NULLIF(@Intensity13, ''),
    Intensity14 = NULLIF(@Intensity14, ''),
    Intensity15 = NULLIF(@Intensity15, ''),
    Intensity16 = NULLIF(@Intensity16, ''),
    Intensity17 = NULLIF(@Intensity17, ''),
    Intensity18 = NULLIF(@Intensity18, ''),
    Intensity19 = NULLIF(@Intensity19, ''),
    Intensity20 = NULLIF(@Intensity20, ''),
    Intensity21 = NULLIF(@Intensity21, ''),
    Intensity22 = NULLIF(@Intensity22, ''),
    Intensity23 = NULLIF(@Intensity23, ''),
    Intensity24 = NULLIF(@Intensity24, ''),
    Intensity25 = NULLIF(@Intensity25, ''),
    Intensity26 = NULLIF(@Intensity26, ''),
    Intensity27 = NULLIF(@Intensity27, ''),
    Intensity28 = NULLIF(@Intensity28, ''),
    Intensity29 = NULLIF(@Intensity29, ''),
    Intensity30 = NULLIF(@Intensity30, ''),
    Intensity31 = NULLIF(@Intensity31, ''),
    Intensity32 = NULLIF(@Intensity32, ''),
    Intensity33 = NULLIF(@Intensity33, ''),
    Intensity34 = NULLIF(@Intensity34, ''),
    Intensity35 = NULLIF(@Intensity35, ''),
    Intensity36 = NULLIF(@Intensity36, ''),
    Intensity37 = NULLIF(@Intensity37, ''),
    Intensity38 = NULLIF(@Intensity38, ''),
    Intensity39 = NULLIF(@Intensity39, ''),
    Intensity40 = NULLIF(@Intensity40, ''),
    Intensity41 = NULLIF(@Intensity41, ''),
    Intensity42 = NULLIF(@Intensity42, ''),
    Intensity43 = NULLIF(@Intensity43, ''),
    Intensity44 = NULLIF(@Intensity44, ''),
    Intensity45 = NULLIF(@Intensity45, ''),
    Intensity46 = NULLIF(@Intensity46, ''),
    Intensity47 = NULLIF(@Intensity47, ''),
    Intensity48 = NULLIF(@Intensity48, ''),
    Intensity49 = NULLIF(@Intensity49, ''),
    Intensity50 = NULLIF(@Intensity50, ''),
    Intensity51 = NULLIF(@Intensity51, ''),
    Intensity52 = NULLIF(@Intensity52, ''),
    Intensity53 = NULLIF(@Intensity53, ''),
    Intensity54 = NULLIF(@Intensity54, ''),
    Intensity55 = NULLIF(@Intensity55, ''),
    Intensity56 = NULLIF(@Intensity56, ''),
    Intensity57 = NULLIF(@Intensity57, ''),
    Intensity58 = NULLIF(@Intensity58, ''),
    Intensity59 = NULLIF(@Intensity59, '');
```

13. minutemetsnarrow

```
CREATE TABLE minutemetsnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    METs INT
);
```

```
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteMETsNarrow_merged.csv"
INTO TABLE minutemetsnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @METs)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    METs = NULLIF(@METs, '');
```

14. Minutesleep

```
CREATE TABLE minutesleep (
    Id BIGINT,
    Date DATETIME,
    Value INT,
    LogId BIGINT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteSleep_merged.csv"
INTO TABLE minutesleep
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Date, @Value, @LogId)
SET
    Id = NULLIF(@Id, ''),
    Date = STR_TO_DATE(@Date, '%m/%d/%Y %r'),
    Value = NULLIF(@Value, ''),
    LogId = NULLIF(@LogId, '');
```

15. Minutestepsnarrow

```
CREATE TABLE minutestepsnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    Steps INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteStepsNarrow_merged.csv"
INTO TABLE minutestepsnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Steps)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
```

```
    Steps = NULLIF(@Steps, '');
```

16. minutestepswide

```
CREATE TABLE minutestepswide (
    Id BIGINT,
    ActivityHour DATETIME,
    Steps00 INT,
    Steps01 INT,
    Steps02 INT,
    Steps03 INT,
    Steps04 INT,
    Steps05 INT,
    Steps06 INT,
    Steps07 INT,
    Steps08 INT,
    Steps09 INT,
    Steps10 INT,
    Steps11 INT,
    Steps12 INT,
    Steps13 INT,
    Steps14 INT,
    Steps15 INT,
    Steps16 INT,
    Steps17 INT,
    Steps18 INT,
    Steps19 INT,
    Steps20 INT,
    Steps21 INT,
    Steps22 INT,
    Steps23 INT,
    Steps24 INT,
    Steps25 INT,
    Steps26 INT,
    Steps27 INT,
    Steps28 INT,
    Steps29 INT,
    Steps30 INT,
    Steps31 INT,
    Steps32 INT,
    Steps33 INT,
    Steps34 INT,
    Steps35 INT,
    Steps36 INT,
    Steps37 INT,
    Steps38 INT,
    Steps39 INT,
    Steps40 INT,
    Steps41 INT,
    Steps42 INT,
    Steps43 INT,
    Steps44 INT,
    Steps45 INT,
    Steps46 INT,
    Steps47 INT,
    Steps48 INT,
    Steps49 INT,
    Steps50 INT,
```

```
    Steps51 INT,
    Steps52 INT,
    Steps53 INT,
    Steps54 INT,
    Steps55 INT,
    Steps56 INT,
    Steps57 INT,
    Steps58 INT,
    Steps59 INT
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteStepsWide_merged.csv"
INTO TABLE minutestepswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour,
 @Steps00, @Steps01, @Steps02, @Steps03, @Steps04, @Steps05, @Steps06,
@Steps07, @Steps08, @Steps09,
 @Steps10, @Steps11, @Steps12, @Steps13, @Steps14, @Steps15, @Steps16,
@Steps17, @Steps18, @Steps19,
 @Steps20, @Steps21, @Steps22, @Steps23, @Steps24, @Steps25, @Steps26,
@Steps27, @Steps28, @Steps29,
 @Steps30, @Steps31, @Steps32, @Steps33, @Steps34, @Steps35, @Steps36,
@Steps37, @Steps38, @Steps39,
 @Steps40, @Steps41, @Steps42, @Steps43, @Steps44, @Steps45, @Steps46,
@Steps47, @Steps48, @Steps49,
 @Steps50, @Steps51, @Steps52, @Steps53, @Steps54, @Steps55, @Steps56,
@Steps57, @Steps58, @Steps59)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Steps00 = NULLIF(@Steps00, ''),
    Steps01 = NULLIF(@Steps01, ''),
    Steps02 = NULLIF(@Steps02, ''),
    Steps03 = NULLIF(@Steps03, ''),
    Steps04 = NULLIF(@Steps04, ''),
    Steps05 = NULLIF(@Steps05, ''),
    Steps06 = NULLIF(@Steps06, ''),
    Steps07 = NULLIF(@Steps07, ''),
    Steps08 = NULLIF(@Steps08, ''),
    Steps09 = NULLIF(@Steps09, ''),
    Steps10 = NULLIF(@Steps10, ''),
    Steps11 = NULLIF(@Steps11, ''),
    Steps12 = NULLIF(@Steps12, ''),
    Steps13 = NULLIF(@Steps13, ''),
    Steps14 = NULLIF(@Steps14, ''),
    Steps15 = NULLIF(@Steps15, ''),
    Steps16 = NULLIF(@Steps16, ''),
    Steps17 = NULLIF(@Steps17, ''),
    Steps18 = NULLIF(@Steps18, ''),
    Steps19 = NULLIF(@Steps19, ''),
    Steps20 = NULLIF(@Steps20, ''),
    Steps21 = NULLIF(@Steps21, ''),
    Steps22 = NULLIF(@Steps22, ''),
    Steps23 = NULLIF(@Steps23, ''),
    Steps24 = NULLIF(@Steps24, ''),
```

```
    Steps25 = NULLIF(@Steps25, ''),
    Steps26 = NULLIF(@Steps26, ''),
    Steps27 = NULLIF(@Steps27, ''),
    Steps28 = NULLIF(@Steps28, ''),
    Steps29 = NULLIF(@Steps29, ''),
    Steps30 = NULLIF(@Steps30, ''),
    Steps31 = NULLIF(@Steps31, ''),
    Steps32 = NULLIF(@Steps32, ''),
    Steps33 = NULLIF(@Steps33, ''),
    Steps34 = NULLIF(@Steps34, ''),
    Steps35 = NULLIF(@Steps35, ''),
    Steps36 = NULLIF(@Steps36, ''),
    Steps37 = NULLIF(@Steps37, ''),
    Steps38 = NULLIF(@Steps38, ''),
    Steps39 = NULLIF(@Steps39, ''),
    Steps40 = NULLIF(@Steps40, ''),
    Steps41 = NULLIF(@Steps41, ''),
    Steps42 = NULLIF(@Steps42, ''),
    Steps43 = NULLIF(@Steps43, ''),
    Steps44 = NULLIF(@Steps44, ''),
    Steps45 = NULLIF(@Steps45, ''),
    Steps46 = NULLIF(@Steps46, ''),
    Steps47 = NULLIF(@Steps47, ''),
    Steps48 = NULLIF(@Steps48, ''),
    Steps49 = NULLIF(@Steps49, ''),
    Steps50 = NULLIF(@Steps50, ''),
    Steps51 = NULLIF(@Steps51, ''),
    Steps52 = NULLIF(@Steps52, ''),
    Steps53 = NULLIF(@Steps53, ''),
    Steps54 = NULLIF(@Steps54, ''),
    Steps55 = NULLIF(@Steps55, ''),
    Steps56 = NULLIF(@Steps56, ''),
    Steps57 = NULLIF(@Steps57, ''),
    Steps58 = NULLIF(@Steps58, ''),
    Steps59 = NULLIF(@Steps59, '');
```

17. sleepday

```
CREATE TABLE sleepday (
    Id BIGINT,
    SleepDay DATETIME,
    TotalSleepRecords INT,
    TotalMinutesAsleep INT,
    TotalTimeInBed INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/sleepDay_merged.csv"
INTO TABLE sleepday
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @SleepDay, @TotalSleepRecords, @TotalMinutesAsleep,
@TotalTimeInBed)
SET
    Id = NULLIF(@Id, ''),
    SleepDay = STR_TO_DATE(@SleepDay, '%m/%d/%Y %r'),
```

```
    TotalSleepRecords = NULLIF(@TotalSleepRecords, ''),
    TotalMinutesAsleep = NULLIF(@TotalMinutesAsleep, ''),
    TotalTimeInBed = NULLIF(@TotalTimeInBed, '');
```

18. weightloginfo

```
CREATE TABLE weightloginfo (
    Id BIGINT,
    Date DATETIME,
    WeightKg DOUBLE,
    WeightPounds DOUBLE,
    Fat BIGINT,
    BMI DOUBLE,
    IsManualReport BOOLEAN,
    LogId BIGINT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/weightLogInfo_merged.csv"
INTO TABLE weightloginfo
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Date, @WeightKg, @WeightPounds, @Fat, @BMI, @IsManualReport,
@LogId)
SET
    Id = NULLIF(@Id, ''),
    Date = STR_TO_DATE(@Date, '%m/%d/%Y %r'),
    WeightKg = NULLIF(@WeightKg, ''),
      WeightPounds = NULLIF(@WeightPounds, ''),
    Fat = NULLIF(@Fat, ''),
    BMI = NULLIF(@BMI, ''),
    IsManualReport = IF(@IsManualReport = 'TRUE', TRUE, FALSE),
    LogId = NULLIF(@LogId, '');
```

MERGE DAILY DATA INTO ONE TABLE

```
CREATE TABLE daily_merged_data AS
SELECT
     -- dailyactivity
    da.Id,
    da.ActivityDate,
    da.TotalSteps,
    da.TotalDistance,
    da.TrackerDistance,
    da.LoggedActivitiesDistance,
    da.VeryActiveDistance,
    da.ModeratelyActiveDistance,
    da.LightActiveDistance,
    da.SedentaryActiveDistance,
    da.VeryActiveMinutes,
    da.FairlyActiveMinutes,
    da.LightlyActiveMinutes,
    da.SedentaryMinutes,
    da.Calories AS activity_calories,

    -- Daily Calories
```

```
    dc.Calories AS daily_calories,

    -- Daily Intensities
    di.SedentaryMinutes AS di_SedentaryMinutes,
    di.LightlyActiveMinutes AS di_LightlyActiveMinutes,
    di.FairlyActiveMinutes AS di_FairlyActiveMinutes,
    di.VeryActiveMinutes AS di_VeryActiveMinutes,
    di.SedentaryActiveDistance AS di_SedentaryActiveDistance,
    di.LightActiveDistance AS di_LightActiveDistance,
    di.ModeratelyActiveDistance AS di_ModeratelyActiveDistance,
    di.VeryActiveDistance AS di_VeryActiveDistance,

    -- Daily Steps
    ds.StepTotal AS ds_step_total,

    -- Heart Rate (daily average)
    hr.avg_heart_rate,

    -- Sleep Day
    sl.TotalSleepRecords,
    sl.TotalMinutesAsleep,
    sl.TotalTimeInBed,

    -- weight log info
    wl.WeightKg,
    wl.WeightPounds,
    wl.Fat,
    wl.BMI,
    wl.LogId

FROM dailyactivity AS da

-- JOIN with daily_calories
LEFT JOIN dailycalories AS dc
  ON da.Id = dc.Id AND da.ActivityDate = dc.ActivityDay

-- JOIN with daily_intensities
LEFT JOIN dailyintensities as di
  ON da.Id = di.Id AND da.ActivityDate = di.ActivityDay

-- JOIN with daily_steps
LEFT JOIN dailysteps as ds
  ON da.Id = ds.Id AND da.ActivityDate = ds.ActivityDay

-- JOIN with average daily heart rate
LEFT JOIN (
    SELECT
        Id,
        DATE(Time) AS HR_Date,
        ROUND(AVG(Value), 2) AS avg_heart_rate
    FROM heartrateseconds
    GROUP BY Id, HR_Date
) as hr
  ON da.Id = hr.Id AND da.ActivityDate = hr.HR_Date

-- JOIN with sleep data
LEFT JOIN sleepday as sl
  ON da.Id = sl.Id AND da.ActivityDate = date(sl.SleepDay)
```

```
-- JOIN with weight data
LEFT JOIN (
    SELECT
        Id,
        DATE(Date) AS WeightDate,
        WeightKg,
        WeightPounds,
        Fat,
        BMI,
        LogId
    FROM weightloginfo
) AS wl
  ON da.Id = wl.Id AND da.ActivityDate = wl.WeightDate;

-- MERGE TIME BASED DATA INTO ANOTHER TABLE

CREATE TABLE timebased_merged_data AS
SELECT
    -- hourly calories
    hc.Id,
    hc.ActivityHour AS TimeStamp,
      hc.Calories AS hourly_calories,

    -- hourly intensities
    hi.TotalIntensity AS hourly_total_intensity,
    hi.AverageIntensity AS hourly_avg_intensity,

    -- hourly steps
    hs.StepTotal AS hourly_steps,

    -- minute calories narrow
    mcn.Calories AS minute_calories_narrow,

    -- minute calories wide
     mcw.Calories00 AS calories_wide00,
       mcw.Calories01 AS calories_wide01,
       mcw.Calories02 AS calories_wide02,
       mcw.Calories03 AS calories_wide03,
       mcw.Calories04 AS calories_wide04,
     mcw.Calories05 AS calories_wide05,
     mcw.Calories06 AS calories_wide06,
     mcw.Calories07 AS calories_wide07,
     mcw.Calories08 AS calories_wide08,
     mcw.Calories09 AS calories_wide09,
     mcw.Calories10 AS calories_wide10,
     mcw.Calories11 AS calories_wide11,
     mcw.Calories12 AS calories_wide12,
     mcw.Calories13 AS calories_wide13,
     mcw.Calories14 AS calories_wide14,
     mcw.Calories15 AS calories_wide15,
     mcw.Calories16 AS calories_wide16,
     mcw.Calories17 AS calories_wide17,
     mcw.Calories18 AS calories_wide18,
     mcw.Calories19 AS calories_wide19,
     mcw.Calories20 AS calories_wide20,
     mcw.Calories21 AS calories_wide21,
     mcw.Calories22 AS calories_wide22,
```

```
 mcw.Calories23 AS calories_wide23,
 mcw.Calories24 AS calories_wide24,
 mcw.Calories25 AS calories_wide25,
 mcw.Calories26 AS calories_wide26,
   mcw.Calories27 AS calories_wide27,
   mcw.Calories28 AS calories_wide28,
 mcw.Calories29 AS calories_wide29,
   mcw.Calories30 AS calories_wide30,
 mcw.Calories31 AS calories_wide31,
 mcw.Calories32 AS calories_wide32,
 mcw.Calories33 AS calories_wide33,
 mcw.Calories34 AS calories_wide34,
 mcw.Calories35 AS calories_wide35,
   mcw.Calories36 AS calories_wide36,
 mcw.Calories37 AS calories_wide37,
 mcw.Calories38 AS calories_wide38,
 mcw.Calories39 AS calories_wide39,
   mcw.Calories40 AS calories_wide40,
 mcw.Calories41 AS calories_wide41,
 mcw.Calories42 AS calories_wide42,
 mcw.Calories43 AS calories_wide43,
 mcw.Calories44 AS calories_wide44,
 mcw.Calories45 AS calories_wide45,
 mcw.Calories46 AS calories_wide46,
 mcw.Calories47 AS calories_wide47,
 mcw.Calories48 AS calories_wide48,
 mcw.Calories49 AS calories_wide49,
 mcw.Calories50 AS calories_wide50,
 mcw.Calories51 AS calories_wide51,
 mcw.Calories52 AS calories_wide52,
 mcw.Calories53 AS calories_wide53,
 mcw.Calories54 AS calories_wide54,
 mcw.Calories55 AS calories_wide55,
 mcw.Calories56 AS calories_wide56,
 mcw.Calories57 AS calories_wide57,
 mcw.Calories58 AS calories_wide58,
 mcw.Calories59 AS calories_wide59,

 -- minute intensities narrow
 minin.Intensity AS minute_intensity_narrow,

-- minute intensities wide
 miniw.Intensity00 AS intensity_wide00,
 miniw.Intensity01 AS intensity_wide01,
 miniw.Intensity02 AS intensity_wide02,
   miniw.Intensity03 AS intensity_wide03,
 miniw.Intensity04 AS intensity_wide04,
 miniw.Intensity05 AS intensity_wide05,
 miniw.Intensity06 AS intensity_wide06,
 miniw.Intensity07 AS intensity_wide07,
 miniw.Intensity08 AS intensity_wide08,
 miniw.Intensity09 AS intensity_wide09,
 miniw.Intensity10 AS intensity_wide10,
 miniw.Intensity11 AS intensity_wide11,
 miniw.Intensity12 AS intensity_wide12,
 miniw.Intensity13 AS intensity_wide13,
 miniw.Intensity14 AS intensity_wide14,
 miniw.Intensity15 AS intensity_wide15,
```

```
  miniw.Intensity16 AS intensity_wide16,
 miniw.Intensity17 AS intensity_wide17,
 miniw.Intensity18 AS intensity_wide18,
 miniw.Intensity19 AS intensity_wide19,
 miniw.Intensity20 AS intensity_wide20,
 miniw.Intensity21 AS intensity_wide21,
 miniw.Intensity22 AS intensity_wide22,
 miniw.Intensity23 AS intensity_wide23,
 miniw.Intensity24 AS intensity_wide24,
 miniw.Intensity25 AS intensity_wide25,
 miniw.Intensity26 AS intensity_wide26,
 miniw.Intensity27 AS intensity_wide27,
  miniw.Intensity28 AS intensity_wide28,
 miniw.Intensity29 AS intensity_wide29,
 miniw.Intensity30 AS intensity_wide30,
 miniw.Intensity31 AS intensity_wide31,
 miniw.Intensity32 AS intensity_wide32,
 miniw.Intensity33 AS intensity_wide33,
 miniw.Intensity34 AS intensity_wide34,
 miniw.Intensity35 AS intensity_wide35,
 miniw.Intensity36 AS intensity_wide36,
 miniw.Intensity37 AS intensity_wide37,
  miniw.Intensity38 AS intensity_wide38,
 miniw.Intensity39 AS intensity_wide39,
 miniw.Intensity40 AS intensity_wide40,
 miniw.Intensity41 AS intensity_wide41,
 miniw.Intensity42 AS intensity_wide42,
 miniw.Intensity43 AS intensity_wide43,
 miniw.Intensity44 AS intensity_wide44,
 miniw.Intensity45 AS intensity_wide45,
 miniw.Intensity46 AS intensity_wide46,
 miniw.Intensity47 AS intensity_wide47,
 miniw.Intensity48 AS intensity_wide48,
 miniw.Intensity49 AS intensity_wide49,
 miniw.Intensity50 AS intensity_wide50,
 miniw.Intensity51 AS intensity_wide51,
 miniw.Intensity52 AS intensity_wide52,
 miniw.Intensity53 AS intensity_wide53,
 miniw.Intensity54 AS intensity_wide54,
 miniw.Intensity55 AS intensity_wide55,
 miniw.Intensity56 AS intensity_wide56,
 miniw.Intensity57 AS intensity_wide57,
 miniw.Intensity58 AS intensity_wide58,
 miniw.Intensity59 AS intensity_wide59,

 -- minute met snarrow
 mm.METs AS minute_mets,

-- minutesleep
 ms.Value AS minute_sleep_value,

 -- minute steps narrow
  msn.Steps AS minute_steps_narrow,

  -- minute steps wide
 msw.Steps00 AS minute_steps_wide00,
 msw.Steps01 AS minute_steps_wide01,
 msw.Steps02 AS minute_steps_wide02,
```

```
msw.Steps03 AS minute_steps_wide03,
  msw.Steps04 AS minute_steps_wide04,
msw.Steps05 AS minute_steps_wide05,
msw.Steps06 AS minute_steps_wide06,
msw.Steps07 AS minute_steps_wide07,
msw.Steps08 AS minute_steps_wide08,
msw.Steps09 AS minute_steps_wide09,
msw.Steps10 AS minute_steps_wide10,
msw.Steps11 AS minute_steps_wide11,
msw.Steps12 AS minute_steps_wide12,
msw.Steps13 AS minute_steps_wide13,
msw.Steps14 AS minute_steps_wide14,
msw.Steps15 AS minute_steps_wide15,
msw.Steps16 AS minute_steps_wide16,
msw.Steps17 AS minute_steps_wide17,
msw.Steps18 AS minute_steps_wide18,
msw.Steps19 AS minute_steps_wide19,
msw.Steps20 AS minute_steps_wide20,
msw.Steps21 AS minute_steps_wide21,
msw.Steps22 AS minute_steps_wide22,
msw.Steps23 AS minute_steps_wide23,
msw.Steps24 AS minute_steps_wide24,
msw.Steps25 AS minute_steps_wide25,
msw.Steps26 AS minute_steps_wide26,
msw.Steps27 AS minute_steps_wide27,
msw.Steps28 AS minute_steps_wide28,
msw.Steps29 AS minute_steps_wide29,
msw.Steps30 AS minute_steps_wide30,
msw.Steps31 AS minute_steps_wide31,
msw.Steps32 AS minute_steps_wide32,
msw.Steps33 AS minute_steps_wide33,
msw.Steps34 AS minute_steps_wide34,
msw.Steps35 AS minute_steps_wide35,
msw.Steps36 AS minute_steps_wide36,
msw.Steps37 AS minute_steps_wide37,
msw.Steps38 AS minute_steps_wide38,
msw.Steps39 AS minute_steps_wide39,
msw.Steps40 AS minute_steps_wide40,
msw.Steps41 AS minute_steps_wide41,
msw.Steps42 AS minute_steps_wide42,
msw.Steps43 AS minute_steps_wide43,
msw.Steps44 AS minute_steps_wide44,
msw.Steps45 AS minute_steps_wide45,
msw.Steps46 AS minute_steps_wide46,
msw.Steps47 AS minute_steps_wide47,
msw.Steps48 AS minute_steps_wide48,
msw.Steps49 AS minute_steps_wide49,
msw.Steps50 AS minute_steps_wide50,
  msw.Steps51 AS minute_steps_wide51,
msw.Steps52 AS minute_steps_wide52,
msw.Steps53 AS minute_steps_wide53,
msw.Steps54 AS minute_steps_wide54,
msw.Steps56 AS minute_steps_wide56,
msw.Steps57 AS minute_steps_wide57,
msw.Steps58 AS minute_steps_wide58,
msw.Steps59 AS minute_steps_wide59
```

```
FROM hourlycalories as hc

-- JOIN hourly intensities
LEFT JOIN hourlyintensities as hi
    ON hc.Id = hi.Id AND hc.ActivityHour = hi.ActivityHour

-- JOIN hourly steps
LEFT JOIN hourlysteps as hs
    ON hc.Id = hs.Id AND hc.ActivityHour = hs.ActivityHour

-- JOIN minuteCaloriesNarrow
LEFT JOIN minutecaloriesnarrow as mcn
    ON hc.Id = mcn.Id AND hc.ActivityHour = mcn.ActivityMinute

-- JOIN minuteCaloriesWide
LEFT JOIN minutecalorieswide as mcw
    ON hc.Id = mcw.Id AND hc.ActivityHour = mcw.ActivityMinute

-- JOIN minuteIntensitiesNarrow
LEFT JOIN minuteintensitiesnarrow as minin
    ON hc.Id = minin.Id AND hc.ActivityHour = minin.ActivityMinute

-- JOIN minuteIntensitiesWide
LEFT JOIN minuteintensitieswide as miniw
    ON hc.Id = miniw.Id AND hc.ActivityHour = miniw.ActivityHour

-- JOIN minuteMETsNarrow
LEFT JOIN minutemetsnarrow as mm
    ON hc.Id = mm.Id AND hc.ActivityHour = mm.ActivityMinute

-- JOIN minuteSleep
LEFT JOIN minutesleep as ms
    ON hc.Id = ms.Id AND hc.ActivityHour = ms.date

-- JOIN minuteStepsNarrow
LEFT JOIN minutestepsnarrow as msn
    ON hc.Id = msn.Id AND hc.ActivityHour = msn.ActivityMinute

-- JOIN minuteStepsWide
LEFT JOIN minutestepswide as msw
    ON hc.Id = msw.Id AND hc.ActivityHour = msw.ActivityHour;


-- MERGE BOTH THE TABLES INTO SINGLE TABLE

CREATE TABLE StravaFullMergedData AS
SELECT
    -- From daily_merged_data
    dmd.Id,
    dmd.ActivityDate,
    dmd.TotalSteps,
    dmd.TotalDistance,
    dmd.TrackerDistance,
    dmd.LoggedActivitiesDistance,
    dmd.VeryActiveDistance,
    dmd.ModeratelyActiveDistance,
    dmd.LightActiveDistance,
    dmd.SedentaryActiveDistance,
```

```
    dmd.VeryActiveMinutes,
    dmd.FairlyActiveMinutes,
    dmd.LightlyActiveMinutes,
    dmd.SedentaryMinutes,
    dmd.activity_calories,
      dmd.daily_calories,
      dmd.di_SedentaryMinutes,
    dmd.di_LightlyActiveMinutes,
    dmd.di_FairlyActiveMinutes,
    dmd.di_VeryActiveMinutes,
      dmd.di_SedentaryActiveDistance,
    dmd.di_LightActiveDistance,
    dmd.di_ModeratelyActiveDistance,
    dmd.di_VeryActiveDistance,
    dmd.ds_step_total,
    dmd.avg_heart_rate,
    dmd.TotalSleepRecords,
    dmd.TotalMinutesAsleep,
    dmd.TotalTimeInBed,
    dmd.WeightKg,
    dmd.Fat,
    dmd.BMI,

    -- From timebased_merged_data
    tmd.TimeStamp,
    tmd.hourly_calories,
    tmd.hourly_total_intensity,
    tmd.hourly_avg_intensity,
    tmd.hourly_steps,
    tmd.minute_calories_narrow,
    tmd.minute_intensity_narrow,
    tmd.minute_mets,
    tmd.minute_sleep_value,
    tmd.minute_steps_narrow,

    -- Calories Wide Columns (00 to 59)
    tmd.calories_wide00, tmd.calories_wide01, tmd.calories_wide02,
tmd.calories_wide03, tmd.calories_wide04,
    tmd.calories_wide05, tmd.calories_wide06, tmd.calories_wide07,
tmd.calories_wide08, tmd.calories_wide09,
    tmd.calories_wide10, tmd.calories_wide11, tmd.calories_wide12,
tmd.calories_wide13, tmd.calories_wide14,
    tmd.calories_wide15, tmd.calories_wide16, tmd.calories_wide17,
tmd.calories_wide18, tmd.calories_wide19,
    tmd.calories_wide20, tmd.calories_wide21, tmd.calories_wide22,
tmd.calories_wide23, tmd.calories_wide24,
    tmd.calories_wide25, tmd.calories_wide26, tmd.calories_wide27,
tmd.calories_wide28, tmd.calories_wide29,
    tmd.calories_wide30, tmd.calories_wide31, tmd.calories_wide32,
tmd.calories_wide33, tmd.calories_wide34,
    tmd.calories_wide35, tmd.calories_wide36, tmd.calories_wide37,
tmd.calories_wide38, tmd.calories_wide39,
    tmd.calories_wide40, tmd.calories_wide41, tmd.calories_wide42,
tmd.calories_wide43, tmd.calories_wide44,
    tmd.calories_wide45, tmd.calories_wide46, tmd.calories_wide47,
tmd.calories_wide48, tmd.calories_wide49,
    tmd.calories_wide50, tmd.calories_wide51, tmd.calories_wide52,
tmd.calories_wide53, tmd.calories_wide54,
```

```
    tmd.calories_wide55, tmd.calories_wide56, tmd.calories_wide57,
tmd.calories_wide58, tmd.calories_wide59,

    -- Intensity Wide Columns (00 to 59)
    tmd.intensity_wide00, tmd.intensity_wide01, tmd.intensity_wide02,
tmd.intensity_wide03, tmd.intensity_wide04,
    tmd.intensity_wide05, tmd.intensity_wide06, tmd.intensity_wide07,
tmd.intensity_wide08, tmd.intensity_wide09,
    tmd.intensity_wide10, tmd.intensity_wide11, tmd.intensity_wide12,
tmd.intensity_wide13, tmd.intensity_wide14,
    tmd.intensity_wide15, tmd.intensity_wide16, tmd.intensity_wide17,
tmd.intensity_wide18, tmd.intensity_wide19,
    tmd.intensity_wide20, tmd.intensity_wide21, tmd.intensity_wide22,
tmd.intensity_wide23, tmd.intensity_wide24,
    tmd.intensity_wide25, tmd.intensity_wide26, tmd.intensity_wide27,
tmd.intensity_wide28, tmd.intensity_wide29,
    tmd.intensity_wide30, tmd.intensity_wide31, tmd.intensity_wide32,
tmd.intensity_wide33, tmd.intensity_wide34,
    tmd.intensity_wide35, tmd.intensity_wide36, tmd.intensity_wide37,
tmd.intensity_wide38, tmd.intensity_wide39,
    tmd.intensity_wide40, tmd.intensity_wide41, tmd.intensity_wide42,
tmd.intensity_wide43, tmd.intensity_wide44,
    tmd.intensity_wide45, tmd.intensity_wide46, tmd.intensity_wide47,
tmd.intensity_wide48, tmd.intensity_wide49,
    tmd.intensity_wide50, tmd.intensity_wide51, tmd.intensity_wide52,
tmd.intensity_wide53, tmd.intensity_wide54,
    tmd.intensity_wide55, tmd.intensity_wide56, tmd.intensity_wide57,
tmd.intensity_wide58, tmd.intensity_wide59,

    -- Steps Wide Columns (00 to 59)
    tmd.minute_steps_wide00, tmd.minute_steps_wide01,
tmd.minute_steps_wide02, tmd.minute_steps_wide03,
tmd.minute_steps_wide04,
    tmd.minute_steps_wide05, tmd.minute_steps_wide06,
tmd.minute_steps_wide07, tmd.minute_steps_wide08,
tmd.minute_steps_wide09,
    tmd.minute_steps_wide10, tmd.minute_steps_wide11,
tmd.minute_steps_wide12, tmd.minute_steps_wide13,
tmd.minute_steps_wide14,
    tmd.minute_steps_wide15, tmd.minute_steps_wide16,
tmd.minute_steps_wide17, tmd.minute_steps_wide18,
tmd.minute_steps_wide19,
    tmd.minute_steps_wide20, tmd.minute_steps_wide21,
tmd.minute_steps_wide22, tmd.minute_steps_wide23,
tmd.minute_steps_wide24,
    tmd.minute_steps_wide25, tmd.minute_steps_wide26,
tmd.minute_steps_wide27, tmd.minute_steps_wide28,
tmd.minute_steps_wide29,
    tmd.minute_steps_wide30, tmd.minute_steps_wide31,
tmd.minute_steps_wide32, tmd.minute_steps_wide33,
tmd.minute_steps_wide34,
    tmd.minute_steps_wide35, tmd.minute_steps_wide36,
tmd.minute_steps_wide37, tmd.minute_steps_wide38,
tmd.minute_steps_wide39,
    tmd.minute_steps_wide40, tmd.minute_steps_wide41,
tmd.minute_steps_wide42, tmd.minute_steps_wide43,
tmd.minute_steps_wide44,
```

```
    tmd.minute_steps_wide45, tmd.minute_steps_wide46,
tmd.minute_steps_wide47, tmd.minute_steps_wide48,
tmd.minute_steps_wide49,
    tmd.minute_steps_wide50, tmd.minute_steps_wide51,
tmd.minute_steps_wide52, tmd.minute_steps_wide53,
tmd.minute_steps_wide54,
    tmd.minute_steps_wide56, tmd.minute_steps_wide57,
tmd.minute_steps_wide58, tmd.minute_steps_wide59

FROM
    daily_merged_data AS dmd
LEFT JOIN
    timebased_merged_data AS tmd
    ON dmd.Id = tmd.Id AND dmd.ActivityDate = DATE(tmd.TimeStamp);
```

PREPROCESSING STEPS

```
-- check datatype
 DESCRIBE stravafullmergeddata;

-- Handle missing values

SET SQL_SAFE_UPDATES = 0;

-- avg_heart_rate
UPDATE stravafullmergeddata
SET avg_heart_rate = (
    SELECT ROUND(AVG(avg_heart_rate), 1)
    FROM (SELECT avg_heart_rate FROM stravafullmergeddata WHERE
avg_heart_rate IS NOT NULL) AS t
)
WHERE avg_heart_rate IS NULL;

-- TotalSleepRecords
UPDATE stravafullmergeddata
SET TotalSleepRecords = (
    SELECT ROUND(AVG(TotalSleepRecords), 1)
    FROM (SELECT TotalSleepRecords FROM stravafullmergeddata WHERE
TotalSleepRecords IS NOT NULL) AS t
)
WHERE TotalSleepRecords IS NULL;

-- TotalMinutesAsleep
UPDATE stravafullmergeddata
SET TotalMinutesAsleep = (
    SELECT ROUND(AVG(TotalMinutesAsleep), 1)
    FROM (SELECT TotalMinutesAsleep FROM stravafullmergeddata WHERE
TotalMinutesAsleep IS NOT NULL) AS t
)
WHERE TotalMinutesAsleep IS NULL;

-- TotalTimeInBed
UPDATE stravafullmergeddata
SET TotalTimeInBed = (
    SELECT ROUND(AVG(TotalTimeInBed), 1)
    FROM (SELECT TotalTimeInBed FROM stravafullmergeddata WHERE
TotalTimeInBed IS NOT NULL) AS t
```

```
)
WHERE TotalTimeInBed IS NULL;

-- WeightKg
UPDATE stravafullmergeddata
SET WeightKg = (
    SELECT ROUND(AVG(WeightKg), 1)
    FROM (SELECT WeightKg FROM stravafullmergeddata WHERE WeightKg IS
NOT NULL) AS t
)
WHERE WeightKg IS NULL;

-- Fat
UPDATE stravafullmergeddata
SET Fat = (
    SELECT ROUND(AVG(Fat), 1)
    FROM (SELECT Fat FROM stravafullmergeddata WHERE Fat IS NOT NULL)
AS t
)
WHERE Fat IS NULL AND Id IS NOT NULL;

-- BMI
UPDATE stravafullmergeddata
SET BMI = (
    SELECT ROUND(AVG(BMI), 2)
    FROM (SELECT BMI FROM stravafullmergeddata WHERE BMI IS NOT NULL)
AS t
)
WHERE BMI IS NULL;

-- minute_sleep_value
UPDATE stravafullmergeddata
SET minute_sleep_value = (
    SELECT ROUND(AVG(minute_sleep_value))
    FROM (SELECT minute_sleep_value FROM stravafullmergeddata WHERE
minute_sleep_value IS NOT NULL) AS t
)
WHERE minute_sleep_value IS NULL AND Id IS NOT NULL;

-- drop missing values
DELETE FROM StravaFullMergedData
WHERE calories_wide00 IS NULL;

# Data Validation
SELECT
    MIN(avg_heart_rate) AS min_hr,
    MAX(avg_heart_rate) AS max_hr,
    MIN(BMI) AS min_bmi,
    MAX(BMI) AS max_bmi
FROM starva.stravafullmergeddata;

-- check for outliners
-- BMI outside normal range
SELECT * FROM stravafullmergeddata
WHERE BMI < 10 OR BMI > 50;

-- Heart rate outliers
SELECT * FROM stravafullmergeddata
```

```
WHERE avg_heart_rate < 30 OR avg_heart_rate > 220;
```

-- Check for Duplicates

```
SELECT Id, ActivityDate, COUNT(*) AS occurrences
FROM stravafullmergeddata
GROUP BY Id, ActivityDate
HAVING COUNT(*) > 1;


SET SQL_SAFE_UPDATES = 0;
DELETE t1
FROM stravafullmergeddata t1
JOIN stravafullmergeddata t2
ON t1.Id = t2.Id
   AND t1.ActivityDate = t2.ActivityDate
   AND t1.temp_row_id > t2.temp_row_id;
ALTER TABLE stravafullmergeddata DROP COLUMN temp_row_id;
```

-- Delete Duplicates

```
ALTER TABLE stravafullmergeddata
ADD COLUMN temp_row_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY;


-- ____________________________________________________________ -
-
```

BUSINESS INSIGHTS

```
-- 1. Daily Activity Table
SELECT ActivityDate FROM dailyactivity;

-- Max Distance By Max Active Distance
SELECT id, max(TotalDistance) AS TopDistance,
max(VeryActiveDistance) AS TopActiveDistance
FROM DailyActivity
GROUP BY id
ORDER BY TopDistance DESC;

-- Top Active Minutes By Calories
SELECT id, max(VeryActiveDistance) AS TopActiveMinutes,
max(TrackerDistance) AS MaxTrackerDistance, Calories
FROM DailyActivity
GROUP BY id, Calories
ORDER BY MaxTrackerDistance DESC, Calories DESC LIMIT 5;

-- Activity Date by Topsteps, TopDistance And Calories
SELECT id, ActivityDate, max(TotalSteps) AS TopSteps,
max(TotalDistance) AS TopDistance,
Calories
FROM DailyActivity
GROUP BY id, ActivityDate, Calories
ORDER BY TopSteps DESC, TopDistance DESC, Calories DESC
LIMIT 10;

-- 2. Daily Calories Table
SELECT * FROM dailycalories;

-- AvgCalories
SELECT ActivityDay, avg(Calories) AS AvgCalories
```

```
FROM dailycalories
GROUP BY ActivityDay;

-- Inner join Daily Activity by total steps
SELECT a.id, a.ActivityDate, a.TotalSteps, c.Calories
FROM dailyactivity a INNER JOIN dailycalories c ON a.id = c.id;

-- Daily Activity Date by Daily Calories (using Window Function)
SELECT ActivityDate, TotalDistance AS MaxDistance, Calories
FROM (
    SELECT a.ActivityDate, a.TotalDistance, c.Calories,
    row_number() OVER (PARTITION BY a.ActivityDate ORDER BY
a.TotalDistance DESC) AS Activity
    FROM dailyactivity a
    INNER JOIN dailycalories c ON a.id = c.id
) sub
ORDER BY MaxDistance DESC
LIMIT 10;

-- Checking for Min & Max values
SELECT
min(TotalDistance), max(TotalDistance), min(TotalSteps),
max(TotalSteps),
min(LoggedActivitiesDistance), max(LoggedActivitiesDistance),
min(SedentaryActiveDistance),
max(SedentaryActiveDistance),
min(SedentaryMinutes), max(SedentaryMinutes),
min(Calories), max(Calories)
FROM dailyactivity;

-- Checking SedentaryMinutes by Daily Patterns
SELECT id, SedentaryMinutes, count(*) AS 'Sedentary Days (=> 1440
mins)'
FROM dailyactivity
WHERE SedentaryMinutes >= 1440
GROUP BY id, SedentaryMinutes;

-- 3. Sleep Day
SELECT * FROM sleepday;

-- Avg Sleep Minutes By Id
SELECT id, avg(TotalMinutesAsleep) AS AvgSleepMinutes
FROM sleepday
GROUP BY id;

-- Sleepday Records more than 2
SELECT id, TotalSleepRecords, count(*) AS MoreSleepRecords
FROM sleepday
WHERE TotalSleepRecords >= 2
GROUP BY id, TotalSleepRecords
ORDER BY TotalSleepRecords DESC;

-- Max Asleep, BedTime by MoreSleepRecords
SELECT id,
max(TotalMinutesAsleep) AS MoreAsleep,
max(TotalTimeInBed) AS MoreBedTime,
max(TotalSleepRecords) AS MoreSleep
FROM sleepday
```

```
GROUP BY id
ORDER BY MoreSleep DESC;

-- 4. Dailysteps
SELECT * FROM dailysteps;

-- Sum Steps By Activity Day
SELECT ActivityDay, sum(StepTotal) AS TotalSteps
FROM dailysteps
GROUP BY ActivityDay;

-- AvgSteps By Id
SELECT id, avg(StepTotal) AS AvgSteps
FROM dailysteps
GROUP BY id;

-- 5. Hourlysteps
SELECT * FROM hourlysteps;

-- Avg Steps BY ActivityHour
SELECT ActivityHour, avg(StepTotal) AS AvgSteps
FROM hourlysteps
GROUP BY ActivityHour;

-- Records where StepTotal is 0
SELECT id, ActivityHour, StepTotal
FROM hourlysteps
WHERE StepTotal = 0
GROUP BY Id, ActivityHour;

-- MaxSteps By ActivityHour Top 10 results
SELECT ActivityHour, max(StepTotal) AS MaxSteps
FROM hourlysteps
GROUP BY ActivityHour
ORDER BY MaxSteps DESC
LIMIT 10;

-- 6. Hourlycalories
SELECT * FROM hourlycalories;

-- AvgCalories by ActivityHour
SELECT ActivityHour, avg(Calories) AS AvgCalories
FROM hourlycalories
GROUP BY ActivityHour
ORDER BY AvgCalories DESC;

-- Top 10 Id by Max Calories
SELECT id, max(Calories) AS MaxCalories
FROM hourlycalories
GROUP BY id
ORDER BY MaxCalories DESC
LIMIT 10;

-- 7. HourlyIntensities
SELECT * FROM hourlyintensities;

-- AvgIntensity by Hour
SELECT hour(ActivityHour) AS Hr, avg(TotalIntensity) AS AvgIntensity
```

```
FROM hourlyintensities
GROUP BY Hr
ORDER BY Hr DESC;

-- MaxIntensity by ActivityHour
SELECT ActivityHour, max(TotalIntensity) AS MaxIntensity
FROM hourlyintensities
GROUP BY ActivityHour
ORDER BY MaxIntensity DESC;

-- 8. Minute Sleep
SELECT * FROM minutesleep;

-- Values By Minute
SELECT minute(date) AS Minutesleep, value
FROM minutesleep
GROUP BY minute(date), value
ORDER BY value DESC;

-- Minute Sleep Where Value is 3
SELECT minute(date) AS Minutesleep, value
FROM minutesleep
WHERE value = 3
GROUP BY minute(date);

-- 9. Heartrate_seconds
-- Avg Heart Rate By Id
SELECT id, avg(Value) AS AvgHeartRate
FROM heartrateseconds
GROUP BY Id;

-- Max Heart Rate by Time (Hr, Minute, Seconds)
SELECT max(Value) AS MaxHeartRate, Hour(Time) AS Hr,
minute(Time) AS Minute, second(Time) AS Seconds
FROM heartrateseconds
GROUP BY Time
ORDER BY MaxHeartRate DESC;

-- 10. Minute Calories
-- AvgCalories by ActivityMinute
SELECT ActivityMinute, avg(Calories) AS AvgCalories
FROM minutecaloriesnarrow
GROUP BY ActivityMinute;

-- MaxCalories by Minute
SELECT minute(ActivityMinute) AS Minute, max(Calories) AS MaxCalories
FROM minuteCaloriesnarrow
GROUP BY minute(ActivityMinute)
ORDER BY MaxCalories DESC;

-- Min Calories by Minute
SELECT minute(ActivityMinute) AS Minute, min(Calories) AS MinCalories
FROM minuteCaloriesnarrow
GROUP BY minute(ActivityMinute)
ORDER BY MinCalories DESC;

-- Top 10 ActivityMinute by TotalCalories
SELECT hour(ActivityMinute) AS Hr,
```

```
minute(ActivityMinute) AS Minute,
second(ActivityMinute) AS Seconds,
sum(Calories) AS TotalCalories
FROM minuteCaloriesnarrow
GROUP BY ActivityMinute
ORDER BY TotalCalories DESC
LIMIT 10;

-- 11. Weight Log Info
SELECT * FROM weightloginfo;

-- AvgWeightKg by AvgBMI
SELECT avg(WeightKg) AS AvgKg, avg(BMI) AS AvgBMI
FROM weightloginfo;

-- IsManualReport By AvgWeightKg and AvgBMI
SELECT IsManualReport, avg(WeightKg) AS AvgKg, avg(BMI) AS AvgBMI,
count(*) AS Count
FROM weightloginfo
GROUP BY IsManualReport
ORDER BY AvgKg DESC;

-- ==============================================
-- SUMMARY DASHBOARD (RECORDS TOTAL)
-- ==============================================
SELECT
(SELECT avg(TotalDistance) FROM dailyactivity) AS AvgDistance,
(SELECT avg(VeryActiveDistance) FROM dailyactivity) AS
VeryActiveDistance,
(SELECT avg(Calories) FROM dailycalories) AS DailyCalories,
(SELECT avg(StepTotal) FROM dailysteps) AS DailySteps,
(SELECT avg(Calories) FROM hourlycalories) AS Hourlycalories,
(SELECT avg(TotalIntensity) FROM hourlyintensities) AS
Hourlyintensities,
(SELECT avg(TotalSleepRecords) FROM sleepday) AS SleepRecord,
(SELECT avg(WeightKg) FROM weightloginfo) AS WeightKg,
(SELECT avg(BMI) FROM weightloginfo) AS BMI,
(SELECT AVG(Value) FROM heartrateseconds) AS HeartRate;

-- -- End of code -- --
```

BUSINESS INSIGHTS RESULTS

1. **Top Performers by Activity**
   • Top Steps, Distance, and Calories:
   [1] Identified users with the highest step counts, longest active distances, and most calories burned.
   [2]. These top performers serve as benchmarks for activity and energy expenditure.

id	ActivityDate	TopSteps	TopDistance	Calories
1624580081	2016-05-01	36019	28.03	2690
8877689391	2016-04-16	29326	25.29	4547
8877689391	2016-04-30	27745	26.72	4398
8877689391	2016-04-27	23629	20.65	3808
8877689391	2016-04-12	23186	20.4	3921
8053475328	2016-04-24	22988	17.95	3577
4388161847	2016-05-07	22770	17.54	4022
8053475328	2016-04-23	22359	17.19	3554
2347167796	2016-04-16	22244	15.08	2670
8053475328	2016-05-08	22026	17.65	3589

2. **Daily Activity Averages**
   • Avg Daily Activities:
   [1]. Calculate average daily steps, calories, and active minutes across all users.
   [2]. Provides a baseline for typical user engagement and activity levels.

AvgDistance	VeryActiveDistance	DailyCalories	DailySteps	Hourlycalories	Hourlyintensities	SleepRecord	WeightKg	BMI	HeartRate
5.489702121915415	1.502680850999945	2303.6096	7637.9106	97.3868	12.0353	1.1186	72.03582137378294	25.185223792915917	77.3284

3. **Heart Rate Analysis**
   • Avg Heart Rate by User:
   [1]. Assessed average heart rate for each user.
   [2]. Revealed individual differences in cardiovascular activity, with some users consistently higher or lower than the group average.

id	AvgHeartRate
2022484408	80.2369
2026352035	93.7763
2347167796	76.7228
4020332650	82.3006
4388161847	66.1330
4558609924	81.6739
5553957443	68.6331
5577150313	69.5645
6117666160	83.7499
6775888955	92.0296
6962181067	77.7240
7007744171	91.1238
8792009665	72.4875
8877689391	83.6112

4. **Sleep Patterns**
   a) Avg Sleep Minutes by User:
   [1]. Summarized average sleep duration per user.
   [2]. Highlighted users with longer or shorter sleep patterns, useful for identifying those who may benefit from sleep interventions.

id	AvgSleepMinutes
1503960366	360.2800
1644430081	294.0000
1844505072	652.0000
1927972279	417.0000
2026352035	506.1786
2320127002	61.0000
2347167796	446.8000
3977333714	293.6429
4020332650	349.3750
4319703577	476.6538
4388161847	403.1250
4445114986	385.1786
4558609924	127.6000

b) Max Asleep & Bedtime (for Users with More Sleep Records):

• Identified users with multiple sleep records and their maximum sleep duration and latest bedtimes.

id	MoreAsleep	MoreBedTime	MoreSleep
1927972279	750	775	3
4388161847	619	641	3
6962181067	630	679	3
1503960366	700	712	2
3977333714	424	626	2
4319703577	692	722	2
4445114986	502	542	2
4702921684	591	612	2
5553957443	775	843	2
5577150313	603	634	2
6117666160	658	698	2
8378563200	611	689	2
1644430081	796	961	1
1844505072	722	961	1
2026352035	573	607	1

5. **Activity Intensity by Time**

• Avg Intensity by Hour:

[1]. Analyzed how activity intensity (steps, calories, active minutes) varies throughout the day.

[2]. Found that activity and calorie burn peak during midday and afternoon hours.

Hr	AvgIntensity
23	4.9967
22	9.0631
21	12.0729
20	14.3400
19	21.3852
18	21.9216
17	21.6556
16	17.7166
15	15.5847
14	18.8686
13	18.7752
12	19.8471
11	16.9213
10	17.6437

6. **Calories & Steps by Day**

a) Avg Calories by Activity Day:

[1]. Compared average calories burned on each day of the week.

[2]. Revealed which days users are most and least active.

ActivityDay	AvgCalories
2016-04-12	2390.6970
2016-04-13	2286.6364
2016-04-14	2356.3939
2016-04-15	2355.1818
2016-04-16	2392.9375
2016-04-17	2230.9688
2016-04-18	2333.3750
2016-04-19	2359.0938
2016-04-20	2395.2188
2016-04-21	2421.8750
2016-04-22	2327.6563
2016-04-23	2397.1563
2016-04-24	2291.4375
2016-04-25	2349.5625

b) Avg Steps by User:

[1]. Ranked users by their average daily step count.

id	AvgSteps
1503960366	12116.7419
1624580081	5743.9032
1644430081	7282.9667
1844505072	2580.0645
1927972279	916.1290
2022484408	11370.6452
2026352035	5566.8710
2320127002	4716.8710
2347167796	9519.6667
2873212765	7555.7742
3372868164	6861.6500
3977333714	10984.5667

7. **Weight & BMI**

a) **Avg Weight and BMI:**

[1]. Calculated average weight and BMI per user.

AvgKg	AvgBMI
72.03582137378294	25.185223792915917

b) **IsManualReport by Weight and BMI:**

[1]. Checked whether weight and BMI values were manually reported, ensuring data reliability.

IsManualReport	AvgKg	AvgBMI	Count
0	87.21923153217023	26.431923132676346	26
1	62.4073173709032	24.39463396770199	41

8. **Sedentary Patterns**

   • **Sedentary Minutes by Users:**

   [1] Analyzed daily sedentary minutes, by assigning 1440 mins condition.

id	SedentaryMinutes	Sedentary Days (=> 1440 mins)
1503960366	1440	1
1844505072	1440	9
1927972279	1440	13
4020332650	1440	14
4057192912	1440	1
4319703577	1440	1
4388161847	1440	1
4702921684	1440	1
5577150313	1440	2
6117666160	1440	5
6290855005	1440	4
6775888955	1440	9

9. Sleep Records

   • Sleepday Records > 2:

   • Identified users with more than two sleep records.

   • Useful for deeper sleep pattern analysis.

id	TotalSleepRecords	MoreSleepRecords
1927972279	3	1
4388161847	3	1
6962181067	3	1
1503960366	2	2
1927972279	2	1
3977333714	2	4
4319703577	2	1
4388161847	2	5
4445114986	2	11
4702921684	2	2
5553957443	2	7

10. **Top 10 Users and Activity Highlights**

    a) Top 10 by Max Calories:

    • Listed the top 10 users with the highest single-day calorie burn.

id	MaxCalories
8877689391	948
8378563200	669
8053475328	651
7086361926	633
5577150313	612
8583815059	597
4388161847	592
6290855005	531
7007744171	493
2022484408	486

b) Top Active Minutes by Calories:

• Highlighted users with the most active minutes and corresponding calories Burned.

id	TopActiveMinutes	MaxTrackerDistance	Calories
1624580081	21.92	28.03	2690
8877689391	21.66	26.72	4398
8877689391	13.24	25.29	4547
8877689391	13.07	20.65	3808
8877689391	12.22	20.4	3921

c) Max Distance by Max Active Distance:

• Showed users with the farthest distances covered in a single day.

id	TopDistance	TopActiveDistance
1624580081	28.03	21.92
8877689391	26.72	21.66
8053475328	17.95	13.4
4388161847	17.54	9.89
2347167796	15.08	5.45
6117666160	15.01	2.03
7007744171	14.3	4.79
1644430081	13.24	4.1
6962181067	13.24	4.28
2022484408	12.91	4.55
8378563200	12.85	7.51
4702921684	12.27	1.82
1503960366	12.21	6.4
8583815059	11.83	6.6
5577150313	11.78	7.65

SUMMARY OF INSIGHTS

- High Performers: A small group of users consistently leads in steps, distance, and calories, setting activity benchmarks.
- Daily and Weekly Patterns: Activity and calorie burn are not evenly distributed; certain days and hours see more engagement.
- Sedentary Behavior: Many users spend significant time sedentary, suggesting opportunities for targeted interventions.
- Sleep and Heart Rate: Variability among users in sleep duration and heart rate highlights the need for personalized recommendations.

- Weight & BMI: Some users fall into higher BMI categories, offering a chance for focused health programs.

BUSINESS RECOMMENDATIONS

- **Personalized "Nudge" Interventions:** Since many users spend the entire day (1,440 minutes) in a sedentary state, the app should implement **smart notifications**. These alerts can prompt users to move during their most inactive hours to improve overall health.
- **Customized Health Coaching:** There is significant variability in sleep and heart rate data across the user base. Moving away from a "one-size-fits-all" approach toward **personalized goals** based on an individual's baseline would provide more value.
- **Targeted Engagement Programs:** Activity levels currently peak during midday and afternoon. To increase daily engagement, the business could launch **early morning or late evening challenges** specifically designed for those off-peak times.
- **Tiered Rewards for High Performers:** A small group of users consistently leads in steps and distance. Creating an **"Elite" tier or leaderboard system** can help retain these power users and turn them into brand ambassadors.
- **Weight Management & BMI Services:** Data shows some users fall into higher BMI categories. This presents an opportunity to offer **premium nutrition plans** or integrated weight-loss tracking features.

CONCLUSION

The Strava Fitness data reveals a diverse user community with distinct habits. While "High Performers" set the bar for activity, a large segment of the population remains sedentary for long periods, indicating a major opportunity for growth and user retention. By transitioning from simple data tracking to **personalized, actionable insights**, the platform can bridge the gap between high-activity users and those needing more motivation, ultimately driving higher user satisfaction and premium subscriptions.